world. Nevertheless, there was a sense in which Veneziano's observation was incomplete. Like memorized formulae used by a student who does not understand their meaning or justification, Euler's beta-function seemed to work, but no one knew why. It was a formula in search of an explanation. This changed in 1970 when the works of Yoichiro Nambu of the University of Chicago, Holger Nielsen of the Niels Bohr Institute, and Leonard Susskind of Stanford University revealed the hitherto-unknown physics lurking behind Euler's formula. These physicists showed that if one modeled elementary particles as little, vibrating, one-dimensional strings, their nuclear interactions could be described exactly by Euler's function. If the pieces of string were small enough, they reasoned, they would still look like point particles, and hence could be consistent with experimental observations.

Although this provided an intuitively simple and pleasing theory, it was not long before the string description of the strong force was shown to fail. During the early 1970s, high-energy experiments capable of probing the subatomic world more deeply showed that the string model made a number of predictions that were in direct conflict with observations. At the same time, the point-particle quantum field theory of quantum chromodynamics was being developed, and its overwhelming success in describing the strong force led to the dismissal of string theory.

Most particle physicists thought that string theory had been relegated to the dustbin of science, but a few dedicated researchers kept at it. Schwarz, for instance, felt that "the mathematical structure of string theory was so beautiful and had so many miraculous properties that it had to be pointing toward something deep."2 One of the problems physicists found with string theory was that it seemed to have a true embarrassment of riches. The theory contained configurations of vibrating string that had properties akin to those of gluons, substantiating its early claim of being a theory of the strong force. But beyond these it contained additional messenger-like particles that did not appear to have any relevance to experimental observations of the strong force. In 1974, Schwarz and Joël Scherk of the Ecole Normale Supérieure made a bold leap that transformed this apparent vice into a virtue. After studying the puzzling messenger-like